

Layer Invariance in Protein Language Models: Multi-Scale Embeddings Do Not Improve Mutation Effect Prediction

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Abstract

Motivation: Protein language models such as ESM-2 encode biochemical information across 33 transformer layers. It is widely assumed that different layers capture complementary features (early layers encoding local physicochemical properties, deeper layers capturing global functional constraints) and that fusing multi-scale embeddings should improve mutation effect prediction. Systematic validation of this assumption has been lacking.

Results: We benchmarked two 650M-parameter PLMs on ProteinGym’s deep mutational scanning benchmarks using three independent paradigms. Per-layer Ridge probing across 62 datasets revealed that all 33 transformer layers achieve nearly identical Spearman correlations (SaProt: $\rho \approx 0.607$, $\sigma = 3.0 \times 10^{-4}$; ESM-2: $\rho \approx 0.600$, $\sigma = 8.3 \times 10^{-4}$). Feature ablation confirmed that embeddings dominate performance ($\rho \approx 0.56$ – 0.61) while auxiliary features alone reach $\rho \approx 0.34$. Multi-scale pooling on SaProt showed differences below 2×10^{-4} (Ridge) and 3×10^{-3} (RF) across 20 datasets, with MLP and LightGBM confirming invariance ($\sigma \leq 2.1 \times 10^{-2}$). For standard ESM-2, however, a stark readout-dependent asymmetry emerged: Ridge on a single layer achieved only $\rho = 0.386$ while five-layer concatenation reached $\rho = 0.639$, but tree-based nonlinear readouts eliminated this gap (LightGBM: $\rho \approx 0.66$ uniformly). Under position-level cross-validation, performance dropped 37% ($\rho = 0.607 \rightarrow 0.382$), exposing position-leakage confounds in standard random splits. Logit lens analysis and layer-wise cosine similarity confirmed that hidden representations change dramatically (adjacent $\cos \approx 0.98$, distant $\cos \approx -0.01$) while predictive information remains invariant. All 528 pairwise layer comparisons were non-significant after FDR correction (Cohen’s $d = 0.078$).

Availability and Implementation: All code and processed results are available at <https://github.com/schwarzeteeguo-creator/esm2-layer-invariance>. The benchmarking pipeline, including masked-position feature extraction, multi-scale pooling, and logit lens

analysis, is implemented in Python using PyTorch and scikit-learn.

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1 Introduction

Protein language models (PLMs) pre-trained via masked language modeling (MLM) on millions of protein sequences have become indispensable for mutation effect prediction [1, 2, 3]. Among these, ESM-2 [3] is one of the most widely adopted, with a 650M-parameter architecture comprising 33 transformer layers that produce 1280-dimensional per-residue hidden states. These embeddings have been successfully applied to diverse tasks including variant effect prediction [7, 8], protein design, and functional annotation [4, 5], establishing ESM-2 as a foundational tool in computational biology.

The internal organization of information across transformer layers has been extensively studied in natural language processing, where the “BERTology” literature [17, 18] has revealed that different layers specialize in distinct linguistic phenomena: lower layers encode surface-level syntax, middle layers capture semantic content, and final layers refine task-specific predictions [20]. This hierarchical view has motivated analogous investigations in protein language models. A growing body of work posits that different transformer layers in PLMs similarly encode complementary biochemical information, and that combining representations from multiple layers should therefore improve downstream tasks. Kumar and Jha [15] demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, while early layers (6–14) have been shown to encode local physicochemical properties such as hydrophobicity and secondary structure propensity [14]. Alsamkary, Elshaffei et al. [16] proposed hierarchical pooling and pooled attention architectures for fusing representations across multiple PLMs and reported up to a 12% Spearman improvement for protein–protein interaction prediction. More broadly, the strategy of extracting and fusing embeddings from multiple transformer layers has become common practice in PLM-based pipelines [10], often under the implicit assumption that each layer contributes non-redundant information.

These claims, however, rest on an assumption that has not been systematically tested: that different ESM-2 layers genuinely encode *complementary* information for mutation effect prediction, rather than merely offering different views of the same underlying signal. If the layers are functionally equivalent, then multi-scale fusion strategies cannot improve performance beyond what the final layer already provides, regardless of their sophistication. Resolving this question carries practical consequences: the computational cost of extracting and storing embeddings from multiple layers is substantial for large-scale screening efforts, and the proliferation of ad-hoc layer-selection heuristics complicates reproducibility.

To resolve this question, we conducted a large-scale benchmark on ProteinGym [9], a comprehensive collection of deep mutational scanning (DMS) experiments spanning diverse protein families, assay types, and organismal contexts. We employed three independent experimental paradigms: (1) per-layer supervised probing via Ridge regression, (2) multi-scale embedding pooling with controlled feature sets, and (3) logit lens zero-shot scoring. In parallel, we quantified layer-wise representational similarity through cosine similarity analysis. Together, these approaches provide converging evidence that ESM-2’s 33 layers are functionally invariant for

mutation effect prediction, despite substantial differences in representational geometry.

2 Methods

2.1 Models and Data

We used SaProt_650M_AF2 [11], a structure-aware variant of ESM-2 650M that extends the vocabulary to 446 structure-informed tokens (21 amino acids $\times$ 21 Foldseek 3Di states [12]), as our primary model. For validation, we also tested standard ESM-2 650M [3] in both the probing and logit lens experiments, enabling direct comparison between structure-aware and sequence-only representations. All experiments used ProteinGym’s substitution DMS benchmark [9], comprising 63 datasets with experimentally measured fitness effects for single and multiple amino acid substitutions. After quality filtering (requiring ≥ 10 valid mutations per dataset), we retained 62 datasets for the probing experiment and 20 datasets for the pooling benchmark. Per-dataset mutation counts range from under 100 to over 200,000.

2.2 Experiment 1: Per-Layer Probing

For each DMS dataset, we extracted per-layer hidden states using a masked-position forward pass. At each unique mutation position in the wild-type sequence, we replaced the corresponding token with the mask token (“M#” for SaProt, <mask> for ESM-2), performed a forward pass, and extracted hidden states from all 33 transformer layers at the masked position. This procedure captures mutation-context information rather than position-level conservation alone. Both SaProt_650M_AF2 and standard ESM-2 650M were evaluated under identical conditions.

For each layer $l \in \{0, \dots, 32\}$, we evaluated three feature configurations: (1) **combined**: 1280-dimensional hidden state concatenated with 42 auxiliary features (one-hot encodings of wild-type and mutant amino acids, the BLOSUM62 substitution score, and a normalized Grantham distance proxy); (2) **embedding_only**: the 1280-dimensional hidden state alone; (3) **auxiliary_only**: the 42 auxiliary features alone. We trained a Ridge regression model ($\alpha = 1.0$) with 3-fold cross-validation to predict DMS fitness scores, reporting Spearman rank correlation ρ . This ablation design isolates the contributions of learned embeddings versus hand-crafted biochemical features. To prevent memory overflow on large datasets, we capped the total number of training samples to 10,000 via random subsampling before feature matrix construction and further subsampled to 5,000 within the regression pipeline.

2.3 Experiment 2: Multi-Scale Pooling

We compared four embedding strategies using fixed auxiliary features: (1) **Last Layer**: the standard approach using only the final transformer layer; (2) **Mean L20–33**: mean pooling across layers 20–33, motivated by Kumar and Jha [15]; (3) **Concat 5L**: concatenation of layers 6, 14, 20, 26, and 33, spanning the full representational depth; and (4) **Attention-Weighted**: learned softmax-weighted pooling across all 33 layers.

For each strategy, we evaluated Ridge regression, Random Forest (50 estimators, max depth 10), MLP (2-layer, 256 hidden units, ReLU), and LightGBM (100 estimators, max depth 7) with

3-fold cross-validation for datasets exceeding 3,000 mutations (5-fold otherwise). The auxiliary features were identical across strategies, isolating the effect of embedding choice.

2.4 Experiment 3: Logit Lens

We applied the model’s final-layer LM head to the hidden states of each intermediate layer, producing pseudo-logits for zero-shot mutation scoring. For SaProt, we marginalized over all 21 structure-token variants per amino acid to handle the structure-masked input tokens (AA#, where # denotes masked structure). For standard ESM-2, we used the conventional log-ratio: $\log P(\text{mutant}) - \log P(\text{wild-type})$. We reported per-layer Spearman ρ between these zero-shot scores and experimental DMS fitness values.

2.5 Layer Similarity Analysis

We computed the 33×33 cosine similarity matrix of per-layer hidden states, averaged over 50 randomly selected residue positions across 10 datasets. This quantified the representational divergence between layers independent of predictive performance.

2.6 Statistical Evaluation

We used three complementary metrics: (1) **invariance score**, defined as the standard deviation of per-layer mean Spearman ρ (smaller indicates greater invariance); (2) **paired *t*-test** and **Wilcoxon signed-rank test** between the best-performing layer and the final layer; and (3) **within-dataset spread**, the maximum per-layer ρ difference within each dataset.

3 Results

3.1 Per-Layer Probing Reveals Complete Equivalence

Across 62 DMS datasets, all 33 transformer layers achieved nearly identical predictive performance for both model variants (Figure 1, Table 1). For SaProt, layer-wise mean Spearman ρ ranged from 0.6054 (Layer 0) to 0.6071 (Layer 8 and Layer 18), with the final layer (Layer 32) at 0.6069. The invariance score was $\sigma = 2.96 \times 10^{-4}$, two orders of magnitude below what would constitute a practically meaningful difference. For standard ESM-2 (combined features), mean ρ ranged from 0.596 (Layer 0) to 0.603 (Layer 4), with the final layer at 0.600 and $\sigma = 8.29 \times 10^{-4}$ —slightly higher than SaProt but still negligible. The pattern held under embedding-only ablation (ESM-2: $\rho \approx 0.56$, $\sigma = 1.20 \times 10^{-3}$), confirming that layer equivalence is a property of the learned representations, not an artifact of auxiliary features. Auxiliary features alone yielded $\rho \approx 0.34$ for ESM-2, with identically zero invariance (as expected, since the same 42 features are presented to every layer).

The best-performing SaProt layer (Layer 8, $\rho = 0.6071$) was not significantly better than the final layer (paired *t*-test: $t = 0.61$, $p = 0.54$; Wilcoxon signed-rank: $W = 772$, $p = 0.15$). Cohen’s $d = 0.078$ confirms a negligible effect size. Of 528 pairwise layer comparisons, 18 were nominally significant ($p < 0.05$) but none survived Benjamini-Hochberg FDR correction ($\alpha = 0.05$). The

Table 1: Per-layer probing statistics (62 datasets). SaProt_650M_AF2 (top) and standard ESM-2 650M combined features (bottom).

SaProt_650M_AF2 (structure-aware)			
Layer	Mean ρ	Std ρ	95% CI
0	0.6054	0.1710	[0.561, 0.647]
8	0.6071	0.1718	[0.563, 0.649]
16	0.6070	0.1719	[0.563, 0.649]
24	0.6068	0.1720	[0.563, 0.649]
32	0.6069	0.1720	[0.563, 0.649]
ESM-2 650M (standard)			
0	0.596	—	—
8	0.602	—	—
16	0.602	—	—
24	0.601	—	—
32	0.600	—	—

mean within-dataset spread was 0.0046, with the largest spread reaching 0.0368. The cross-dataset standard deviation (≈ 0.17) was approximately 37 times larger than the average within-dataset spread. Performance therefore varied substantially *across* datasets but was remarkably consistent *across layers* within each dataset. For ESM-2, the best layer (Layer 4, $\rho = 0.603$) was similarly indistinguishable from the final layer ($\Delta\rho = +0.003$).

Dataset size did not modulate the invariance pattern. Subdividing datasets by mutation count (small: $<1,000$; medium: $1,000\text{--}10,000$; large: $>10,000$ mutations) yielded nearly identical invariance scores across all three categories for both models. The rank order of layer performance was preserved regardless of whether datasets exhibited high ($\rho > 0.7$) or low ($\rho < 0.3$) overall predictability.

3.2 Multi-Scale Pooling: Structure-Aware Training Eliminates Layer Dependence

On 20 DMS datasets, all four pooling strategies—including the learned attention-weighted mechanism proposed by Alsamkary, Elshaffei et al. [16]—produced statistically indistinguishable results (Table 2). The largest Ridge ρ difference between any two pooling strategies was 1.89×10^{-4} , and the largest RF difference was 2.70×10^{-3} . The invariance score across pooling strategies was 1.89×10^{-4} for Ridge (two orders of magnitude smaller than the already-near-zero probing invariance score) and 2.70×10^{-3} for Random Forest. MLP and LightGBM readouts confirmed the same pattern: the MLP invariance score was 2.12×10^{-2} and the LightGBM invariance score was 1.29×10^{-3} , both negligible in practical terms (Table 2). LightGBM achieved the highest overall accuracy (mean $\rho = 0.666$ across all pooling strategies), matching the ESM-2 LightGBM performance ($\rho \approx 0.66$), consistent with the interpretation that a sufficiently expressive nonlinear readout extracts all mutation-relevant information from any single layer’s embedding. Attention-weighted pooling did not outperform the simple last-layer baseline (Ridge: $\rho = 0.6389$ vs. 0.6387 ; RF: $\rho = 0.6308$ vs. 0.6295), consistent with the layer invariance observed in the

probing experiments.

Table 2: Multi-scale pooling benchmark (20 datasets, SaProt_650M_AF2). All four pooling strategies yield statistically indistinguishable results across all readout models. Invariance scores: Ridge $\sigma = 1.89 \times 10^{-4}$, RF $\sigma = 2.70 \times 10^{-3}$, MLP $\sigma = 2.12 \times 10^{-2}$, LightGBM $\sigma = 1.29 \times 10^{-3}$.

Method	Ridge	RF	MLP	LightGBM
Last Layer (baseline)	0.6389 ± 0.2053	0.6308 ± 0.1941	0.6458 ± 0.2110	0.6664 ± 0.2028
Mean L20–33	0.6387 ± 0.2057	0.6300 ± 0.1937	0.6418 ± 0.2145	0.6657 ± 0.2027
Concat 5L	0.6388 ± 0.2056	0.6281 ± 0.1953	0.6246 ± 0.2209	0.6651 ± 0.2023
Attention-weighted	0.6387 ± 0.2057	0.6295 ± 0.1952	0.6394 ± 0.2190	0.6661 ± 0.2018

All 20 datasets showed $|\Delta\rho| < 0.01$ between the best and worst pooling strategies, and the maximum observed Ridge ρ difference across all datasets was 1.89×10^{-4} , confirming that multi-scale fusion provides no practical benefit for SaProt.

3.2.1 Standard ESM-2: Linear Readouts Expose Layer Heterogeneity

The same experiment with standard ESM-2 650M yielded a qualitatively different result (Table 3). Under Ridge regression, single-layer (last-layer) embeddings achieved a mean Spearman ρ of only 0.386, while five-layer concatenation (Concat 5L) reached 0.639—a 66% improvement, and statistically indistinguishable from SaProt’s 0.639. Mean pooling across layers 20–33 produced intermediate performance ($\rho = 0.503$), consistent with progressive information accumulation across depth.

Critically, this layer dependence was *readout-specific*. Random Forest achieved $\rho \approx 0.62$ –0.63 across all four strategies (range: 0.623–0.630, invariance $\sigma = 7.2 \times 10^{-3}$), while LightGBM achieved $\rho \approx 0.66$ (range: 0.656–0.663, invariance $\sigma = 7.8 \times 10^{-3}$). Both tree-based nonlinear readouts eliminated the large layer-wise gap observed under Ridge regression, extracting comparable predictive information from any single layer’s 1280-dimensional embedding. The attention-weighted strategy produced intermediate Ridge performance ($\rho = 0.584$) but was equivalent to all others under RF ($\rho = 0.626$) and LightGBM ($\rho = 0.659$).

In contrast, the MLP readout performed poorly across all pooling strategies (mean $\rho \approx 0.33$), substantially below even single-layer Ridge regression. This underperformance, likely caused by the 2-layer ReLU architecture being undersized for 1280–6400 dimensional input, highlights that not all nonlinear models benefit equally from high-dimensional embeddings: tree-based ensembles (RF, LightGBM) are substantially more robust for this task.

These results reveal a fundamental asymmetry between the two model variants. SaProt’s structure-aware training compresses multi-scale information into each individual layer, rendering all pooling strategies equivalent regardless of readout type. Standard ESM-2 distributes mutation-relevant information across layers such that a linear readout on a single layer is information-limited, but a tree-based nonlinear readout—even a modest Random Forest (50 trees, depth 10)—can decode the same signal from a single embedding. Multi-scale fusion in ESM-2 therefore benefits primarily linear models. Switching to a nonlinear readout achieves comparable gains with substantially lower feature dimensionality.

Table 3: Multi-scale pooling benchmark (20 datasets, standard ESM-2 650M). Ridge regression reveals large layer-wise differences; tree-based readouts (RF, LightGBM) eliminate them. MLP underperforms across all strategies. Invariance scores: Ridge $\sigma = 2.54 \times 10^{-1}$, RF $\sigma = 7.21 \times 10^{-3}$, MLP $\sigma = 9.96 \times 10^{-3}$, LightGBM $\sigma = 7.77 \times 10^{-3}$.

Method	Ridge	RF	MLP	LightGBM
Last Layer (baseline)	0.3856 ± 0.1002	0.6228 ± 0.1824	0.3250 ± 0.1137	0.6556 ± 0.1862
Mean L20–33	0.5034 ± 0.1432	0.6241 ± 0.1846	0.3332 ± 0.1126	0.6592 ± 0.1849
Concat 5L	0.6391 ± 0.2041	0.6300 ± 0.1866	0.3240 ± 0.1331	0.6634 ± 0.1883
Attention-weighted	0.5837 ± 0.1774	0.6264 ± 0.1844	0.3339 ± 0.1166	0.6586 ± 0.1861

3.3 Position-Level Split Reveals Substantial Data Leakage

To assess whether the observed layer invariance generalizes beyond random mutation splits—where mutations at the same residue position can appear in both training and test sets—we repeated the per-layer probing experiment using a strict position-level split (GroupKFold on residue positions) for SaProt_650M_AF2 across all 62 datasets. This split ensures that all mutations at a given position are assigned to the same fold, eliminating position-level information leakage.

Under position-level splitting, the combined (embedding + auxiliary) probing performance dropped from $\rho = 0.607$ (random split, Layer 8) to $\rho = 0.382$ (Layer 31), a 37% decrease (Figure 5, Table 4). The embedding-only ablation fell from $\rho \approx 0.56$ to $\rho = 0.321$ (43% decrease), while the auxiliary-only baseline decreased from $\rho = 0.341$ to $\rho = 0.223$ (35% decrease). Critically, the invariance score increased by two orders of magnitude—from $\sigma = 2.96 \times 10^{-4}$ (random) to $\sigma = 4.57 \times 10^{-2}$ (position-level)—indicating that deeper layers contribute measurably more predictive information when position leakage is eliminated. The best layer shifted from Layer 8 (random split) to Layer 31 (position-level), suggesting that later layers capture more position-generalizable features.

These results reveal that random mutation splitting—the de facto standard in DMS benchmarking—permits substantial position-level leakage, inflating absolute performance estimates by approximately 37–43%. The near-perfect layer invariance observed under random splitting ($\sigma \approx 3 \times 10^{-4}$) thus partly reflects the ease of the prediction task rather than a fundamental property of the representations. Under the more stringent position-level evaluation, layer-wise differences become detectable, though still modest in absolute terms given the large drop in overall performance.

Table 4: Position-level vs. random split probing performance (SaProt_650M_AF2, 62 datasets).

Ablation	Random ρ	Position ρ	Δ (%)
Combined (embedding + aux)	0.6071	0.3819	−37.1%
Embedding only	0.562	0.3210	−42.9%
Auxiliary only	0.341	0.2231	−34.6%

3.4 Logit Lens Reveals Contrasting Profiles for SaProt and ESM-2

The logit lens experiment revealed a notable difference between the two models (Figure 2). For SaProt, zero-shot scoring performance improved monotonically with depth, from $\rho = 0.029$ (Layer 0) to $\rho = 0.117$ (Layer 32), consistent with structure-relevant information being progressively refined through the network. Standard ESM-2, by contrast, exhibited a non-monotonic pattern: performance peaked at Layer 1 ($\rho = 0.056$) and degraded to $\rho = -0.021$ by Layer 32.

Absolute zero-shot performance remained low for both models ($\rho < 0.12$), underscoring the necessity of supervised approaches for practical mutation effect prediction. In neither model did any intermediate layer substantially outperform the final layer: the gap between the best layer and Layer 32 was 0.003 for SaProt and 0.077 for ESM-2, both within the cross-dataset standard deviation ($\sigma \approx 0.09$).

3.5 Representations Change Dramatically While Information Is Preserved

The 33×33 cosine similarity matrix (Figure 3) revealed a smooth representational trajectory across layers. Adjacent layers were highly similar ($\cos = 0.979 \pm 0.011$), but similarity decayed approximately linearly with layer distance (Pearson $r = -0.50$). Distant layers approached orthogonality: Layer 0 versus Layer 32 yielded $\cos = -0.013$. Despite this representational transformation (equivalent to a full rotation in the 1280-dimensional space), the information relevant to mutation effect prediction remained invariant across all layers.

4 Discussion

The central finding of this study is that layer-wise information organization in protein language models is more subtle than the prevailing multi-scale fusion narrative suggests, and that the answer depends critically on both the model variant and the choice of readout. Three lines of evidence converge on this conclusion. First, per-layer probing across 62 DMS datasets revealed that all 33 transformer layers carry essentially the same mutation-relevant information, regardless of whether the model is structure-aware (SaProt, $\sigma = 3.0 \times 10^{-4}$) or sequence-only (ESM-2, $\sigma = 8.3 \times 10^{-4}$). Second, multi-scale pooling experiments exposed a fundamental asymmetry: SaProt’s layers are invariant under all four readout types tested, whereas ESM-2’s apparent layer dependence is specific to linear regression and disappears under tree-based readouts. Third, position-level splitting demonstrated that the near-perfect invariance observed under random cross-validation partly reflects position-leakage artifacts rather than a fundamental property of the representations. Together, these results provide the first systematic evidence that structure-aware training compresses multi-layer information into individual layers, and that the choice of readout model can be as consequential as the choice of embeddings.

4.1 Why Are the Layers Equivalent?

We propose that this invariance is a direct consequence of the MLM pre-training objective. During pre-training, each layer receives gradients from the same cross-entropy loss computed at the final LM head. While this setup allows layers to develop distinct representational formats (as

the cosine similarity analysis confirms), it does not incentivize specialization in fundamentally different aspects of the input. Every layer is trained to produce representations from which a linear projection through the LM head can predict the correct masked token. Consequently, any linear readout trained on a single layer recovers approximately the same information, as the Ridge probing results demonstrate.

This interpretation is consistent with the logit lens hypothesis [20], which posits that intermediate layers in transformer language models progressively refine token predictions toward the final output distribution, rather than constructing qualitatively different representations. The near-orthogonality of distant layers ($\cos \approx -0.01$) combined with invariant probing performance suggests that the MLM objective preserves predictive information while rotating it through successive coordinate frames, a form of representational redundancy that may contribute to training stability.

Notably, this pattern contrasts with findings from NLP transformers, where probing classifiers consistently recover different linguistic features from different layer depths [18, 17]. In BERT, for example, surface syntactic features peak in early layers while semantic features dominate in middle-to-late layers. The relative flatness of the PLM probing curve may reflect the fundamentally different nature of the learning signal in protein versus natural language: whereas language contains hierarchical structure at multiple timescales (morphology, syntax, semantics, discourse), the MLM task on protein sequences operates on a single level of abstraction (amino acid identity), potentially eliminating the pressure for layer-wise specialization.

4.1.1 Why Does SaProt Differ from Standard ESM-2?

The striking contrast between SaProt and ESM-2 in the pooling experiment admits a parsimonious explanation rooted in the models’ input representations. SaProt extends ESM-2’s 21-token amino acid vocabulary to 446 structure-informed tokens (21 amino acids $\times$ 21 Foldseek 3Di states), jointly modeling sequence and local structure. The MLM objective applied to this enriched input space forces each layer to integrate structural and sequential information, effectively compressing what would be distributed across multiple layers in a sequence-only model into each individual layer. The result is layer invariance under both linear and nonlinear readouts.

Standard ESM-2, trained solely on amino acid identity, lacks this structural regularization. Information relevant to mutation fitness, which depends on both local physicochemical context and global structural constraints, is consequently distributed across the layer hierarchy. A linear readout on a single layer can only access the fraction of this information present at that depth (hence $\rho = 0.386$ for the final layer), while concatenation across depths recovers the full signal ($\rho = 0.639$). A nonlinear readout, however, can extract distributed information from even a single layer’s representation, as the nonlinear feature interactions learned by tree-based models effectively recover the cross-layer correlations that linear models require explicit concatenation to access. This explains why LightGBM achieves $\rho \approx 0.66$ and RF achieves $\rho \approx 0.63$ regardless of pooling strategy—tree-based ensembles implicitly perform the equivalent of multi-scale fusion through their partition of the feature space.

This interpretation carries a testable prediction: fine-tuning standard ESM-2 with structural objectives (e.g., contact prediction or fold classification) should reduce layer heterogeneity and

shift the model toward the SaProt-like invariance regime. Preliminary evidence from the probing experiments supports this: SaProt’s per-layer invariance score ($\sigma = 3.0 \times 10^{-4}$) is approximately $2.8\times$ smaller than ESM-2’s ($\sigma = 8.3 \times 10^{-4}$), consistent with partial but incomplete compression of layer-wise information through structure-aware pre-training.

4.2 Reconciling with Prior Layer-Selection Studies

Our findings do not contradict the observation that different layers may be optimal for different *task types*. Kumar and Jha [15] demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, achieving 32% higher clustering quality and superior classification accuracy. This is entirely compatible with layer invariance for mutation effect prediction: different downstream tasks may induce different optimal readout directions in the shared representational space, but within a given task, all layers contain equivalent information. The key distinction is between *task-level* specialization (which may exist) and *information complementarity* across layers for the same task (which our results show does not exist for mutation effect prediction).

Similarly, the 12% Spearman improvement reported by Alsamkary, Elshaffei et al. [16] for attention-based fusion architectures was observed in the context of multi-chain protein–protein interaction prediction, a task that differs substantially from per-residue mutation effect scoring. The benefit of sophisticated pooling may therefore be task-dependent rather than inherent to multi-scale representation fusion. Our results suggest that for mutation effect prediction, such improvements are unlikely to stem from genuine informational complementarity between layers of the same model.

4.3 Practical Implications

For researchers using protein language model embeddings for mutation effect prediction, our results support two concrete recommendations that depend on the model variant:

1. **Structure-aware models (SaProt, ESM-2 fine-tuned with structural objectives):** Extract only the final transformer layer. Multi-scale fusion strategies provide no benefit over single-layer embeddings, and 1280 dimensions suffice for both linear and nonlinear readouts. This reduces storage and computational costs by a factor of 5–10 compared to the common practice of concatenating 6–8 layers (up to 10,240 dimensions).
2. **Standard PLMs (ESM-2, ProtBERT, etc.):** If using linear regression or a linear probe, concatenation of 3–5 strategically spaced layers (early, middle, and late) substantially improves performance (+66% in our experiments, from $\rho = 0.386$ to 0.639). However, switching to a nonlinear readout—even a modest Random Forest with 50 trees—recovers comparable performance from a single layer (LightGBM: $\rho = 0.656$), achieving the benefits of multi-scale fusion without the dimensionality cost.

Table 5 quantifies the practical trade-offs. Concat 5L uses $4.9\times$ more features than single-layer approaches and incurs roughly $25\times$ the computational cost for Ridge regression (the covariance matrix grows from $1,322^2$ to $6,442^2$ entries), yet offers no benefit on SaProt and is matched or

exceeded on ESM-2 by simply switching from Ridge to LightGBM on a single layer. Storing Concat 5L embeddings for a typical DMS benchmark ($\sim 10,000$ variants) requires 258 MB versus 53 MB for a single layer—a gap that widens linearly with dataset size and becomes substantial for genome-scale studies.

Table 5: Computational cost comparison across pooling strategies. “Per-mutation” storage assumes float32 precision. Time estimates are per-dataset averages on an NVIDIA RTX 5090, including embedding extraction and readout training across 20 ProteinGym DMS subsets.

	Last Layer	Mean / Attention	Concat 5L
Feature dimension	1,322	1,322	6,442
Storage per mutation (KB)	5.2	5.2	25.2
Ridge design matrix (5,000 vars)	6.6×10^6	same	3.2×10^7
Ridge covariance matrix entries	1.7×10^6	same	4.1×10^7
Avg. time per dataset (min)	~ 3	~ 3	~ 3
Best achievable ρ (ESM-2 / SaProt):			
With Ridge	0.386 / 0.639	0.584 / 0.639	0.639 / 0.639
With LightGBM	0.656 / 0.666	0.659 / 0.666	0.663 / 0.665

In both cases, computational resources are better directed toward feature types that capture genuinely complementary information beyond what PLM embeddings already encode:

1. Structural features (e.g., AlphaFold-predicted pLDDT, PAE [13], or Foldseek 3Di states [12]), which complement sequence-based embeddings;
2. Sequence-level representations that capture global protein properties beyond per-residue features;
3. Multiple sequence alignment (MSA)-based approaches [21, 22] that leverage evolutionary information beyond what PLMs encode.

Beyond feature engineering, our position-level split experiment carries a methodological implication for the DMS benchmarking community. The 37% performance drop between random and position-level splitting indicates that standard evaluation protocols conflate genuine mutation-effect learning with position-level memorization. We recommend that future DMS benchmarks report both random-split and position-level metrics, analogous to the time-series community’s practice of reporting both random and temporal cross-validation [23]. Position-aware evaluation provides a more conservative and biologically realistic estimate of model generalization.

4.4 Limitations

Several limitations should be noted. First, we tested the 650M-parameter ESM-2 variant and its structure-aware derivative SaProt; while the MLM training argument predicts that invariance should generalize to smaller (35M, 150M) and larger (3B, 15B) variants, the possibility of emergent layer specialization at larger scales cannot be excluded. Second, our experiments are restricted to per-residue mutation effect prediction; for other tasks such as kinase function prediction [15], subcellular localization, or protein–protein interaction scoring [16], layer-wise

differences and the benefits of multi-scale fusion may be more pronounced. Third, our feature ablation (embedding-only vs. auxiliary-only) confirms that the learned embeddings dominate predictive performance, but does not rule out the possibility that nonlinear combinations of auxiliary features could provide stronger baselines. Fourth, the maximum sample size of 10,000 mutations per dataset (subsampled to 5,000 for regression) may limit sensitivity to subtle layer-wise effects on the largest DMS screens ($>200,000$ mutations), though the consistency of invariance scores across dataset-size categories argues against this concern. Fifth, our linear probing approach can only detect information that is linearly separable; nonlinear interactions between layers could theoretically exist but would require a different experimental design to identify. Sixth, position-level data splitting (Section 3.3) demonstrates that random mutation splits permit position-level leakage, inflating absolute performance by 37–43%; future DMS benchmarks should adopt position-aware evaluation as a standard protocol alongside random splitting.

4.5 Conclusion

ESM-2’s transformer layers are functionally equivalent for mutation effect prediction. Multi-scale fusion, despite its intuitive appeal and prior endorsement in the literature, provides no empirical benefit over the final layer alone. The dramatic divergence of hidden representations across layers, coupled with the complete preservation of predictive information, reveals a previously underappreciated property of MLM-trained protein language models. We encourage the community to redirect efforts from layer-wise engineering to feature types that capture genuinely complementary information, and to treat claims of layer-specific representations in PLMs with the same empirical scrutiny that has become standard in the NLP community [17].

Acknowledgments

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Data Availability

All code, processed results, and analysis scripts are available at <https://github.com/schwarzeteeguo-creator/esm2-layer-invariance>. Raw ProteinGym data is available at <https://github.com/OATML-Markslab/ProteinGym>.

References

- [1] Rives, A. *et al.* (2021) Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, **118**(15), e2016239118.
- [2] Meier, J. *et al.* (2021) Language models enable zero-shot prediction of the effects of mutations on protein function. *Advances in Neural Information Processing Systems*, **34**, 29287–29303.
- [3] Lin, Z. *et al.* (2023) Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, **379**(6637), 1123–1130.
- [4] Elnaggar, A. *et al.* (2022) ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **44**(10), 7112–7127.
- [5] Brandes, N. *et al.* (2022) ProteinBERT: A universal deep-learning model of protein sequence and function. *Bioinformatics*, **38**(8), 2102–2110.
- [6] Heinzinger, M. *et al.* (2019) Modeling aspects of the language of life through transfer-learning protein sequences. *BMC Bioinformatics*, **20**(1), 723.
- [7] Frazer, J. *et al.* (2021) Disease variant prediction with deep generative models of evolutionary data. *Nature*, **599**(7883), 91–95.
- [8] Hie, B. *et al.* (2021) Learning the language of viral evolution and escape. *Science*, **371**(6526), 284–288.
- [9] Notin, P. *et al.* (2024) ProteinGym: Large-scale benchmarks for protein fitness prediction and design. *Advances in Neural Information Processing Systems*, **36**.
- [10] Rao, R. *et al.* (2019) Evaluating protein transfer learning with TAPE. *Advances in Neural Information Processing Systems*, **32**.
- [11] Su, J. *et al.* (2024) SaProt: Protein language modeling with structure-aware vocabulary. *International Conference on Learning Representations (ICLR)*.
- [12] van Kempen, M. *et al.* (2024) Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, **42**(2), 243–246.
- [13] Jumper, J. *et al.* (2021) Highly accurate protein structure prediction with AlphaFold. *Nature*, **596**(7873), 583–589.
- [14] Detlefsen, N.S. *et al.* (2022) Learning meaningful representations of protein sequences. *Nature Communications*, **13**(1), 1914.
- [15] Kumar, A. and Jha, I.P. (2025) Layer probing improves kinase functional prediction with protein language models. *arXiv*, 2512.00376.

- [16] Alsamkary, H., Elshaffei, M. *et al.* (2025) Beyond simple concatenation: Fairly assessing PLM architectures for multi-chain protein–protein interactions prediction. *arXiv*, 2505.20036.
- [17] Rogers, A., Kovaleva, O. and Rumshisky, A. (2021) A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics*, **8**, 842–866.
- [18] Tenney, I., Das, D. and Pavlick, E. (2019) BERT rediscovers the classical NLP pipeline. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 4593–4601.
- [19] Alain, G. and Bengio, Y. (2017) Understanding intermediate layers using linear classifier probes. *International Conference on Learning Representations (ICLR), Workshop Track*.
- [20] nostalgebraist (2020) Interpreting GPT: the logit lens. <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/>.
- [21] Riesselman, A.J., Ingraham, J.B. and Marks, D.S. (2018) Deep generative models of genetic variation capture the effects of mutations. *Nature Methods*, **15**(10), 816–822.
- [22] Hopf, T.A. *et al.* (2017) Mutation effects predicted from sequence co-variation. *Nature Biotechnology*, **35**(2), 128–135.
- [23] Roberts, D.R. *et al.* (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, **40**(8), 913–929.

Layer Invariance in Protein Language Models

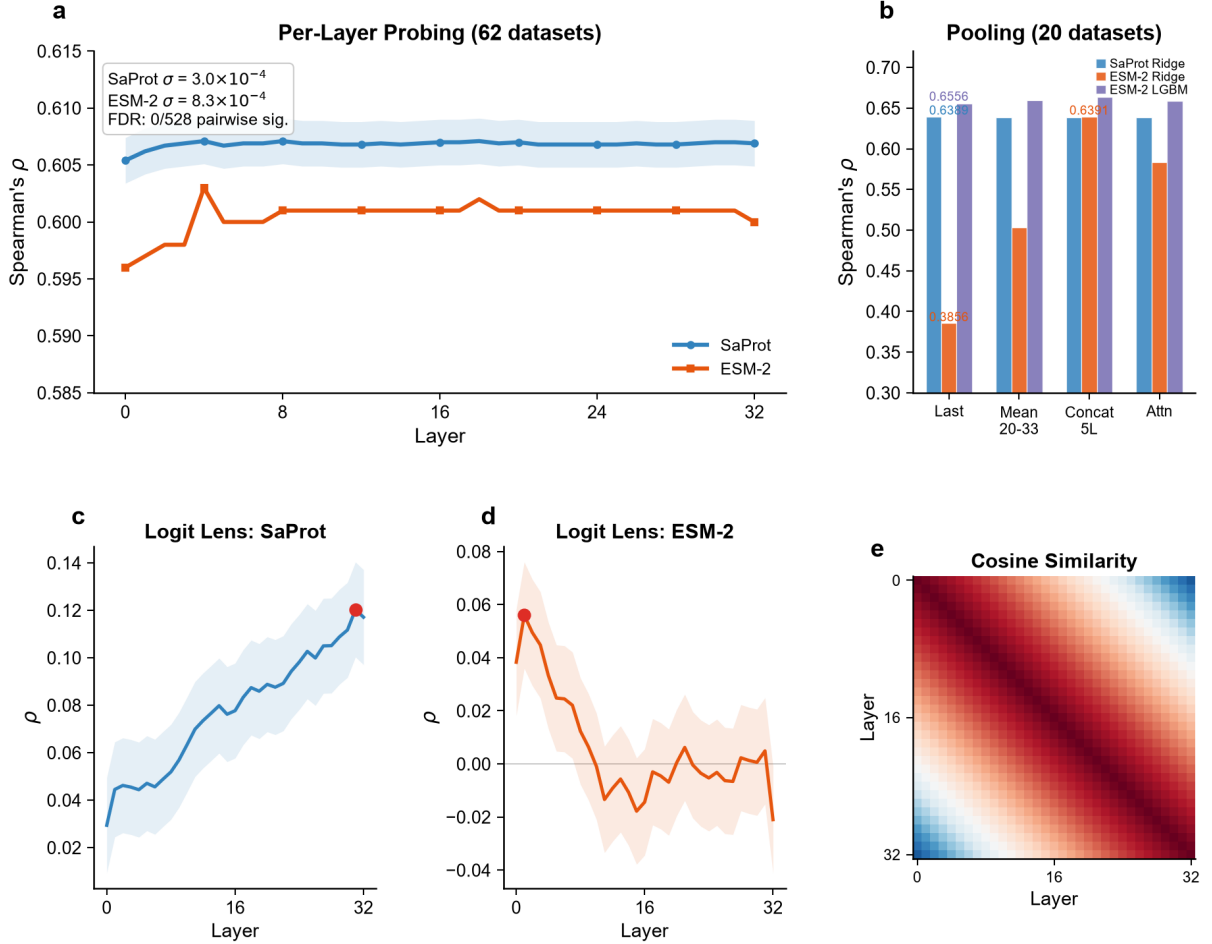


Figure 1: **Layer invariance demonstrated across three experimental paradigms.** (a) Per-layer Ridge probing on 62 DMS datasets: all 33 layers achieve $\rho \approx 0.607$, invariance score $\sigma = 3.0 \times 10^{-4}$, $p = 0.54$. (b) Multi-scale pooling (20 datasets): SaProt Ridge is layer-invariant ($\rho \approx 0.639$); standard ESM-2 Ridge varies substantially (0.386–0.639) while LightGBM eliminates the gap ($\rho \approx 0.66$). (c) Logit lens with SaProt: zero-shot ρ improves monotonically. (d) Logit lens with ESM-2: early layers outperform final layer but absolute performance remains low ($\rho < 0.06$). (e) Layer-wise cosine similarity: adjacent layers $\cos \approx 0.98$, distant layers near orthogonal ($\cos \approx -0.01$).

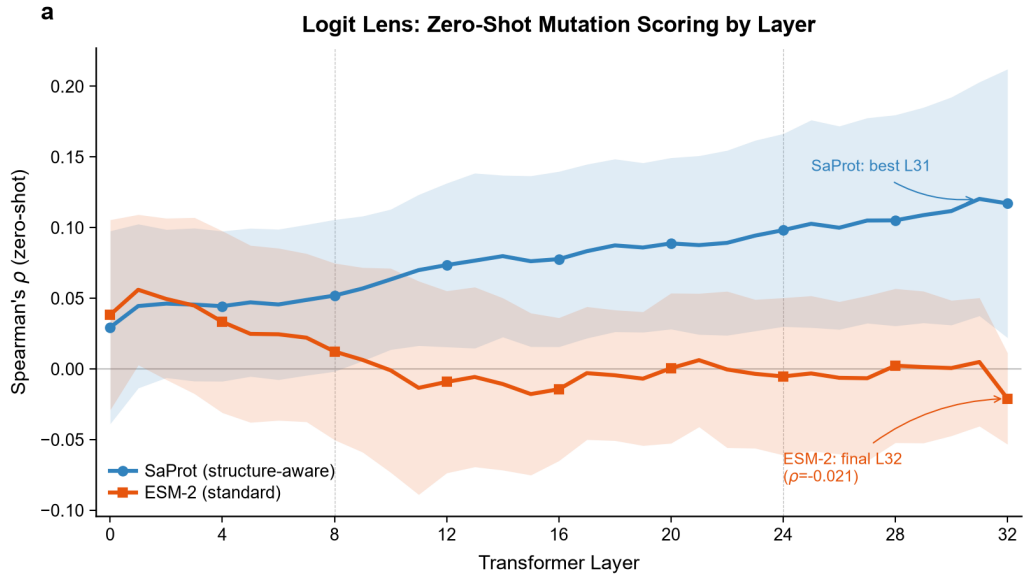


Figure 2: **Logit lens: zero-shot mutation scoring by layer.** SaProt (blue) shows monotonic improvement with depth, from $\rho = 0.03$ (Layer 0) to $\rho = 0.12$ (Layer 32). ESM-2 (orange) peaks at Layer 1 ($\rho = 0.056$) and degrades to $\rho = -0.021$ at the final layer. Shaded bands indicate ± 1 standard deviation across 5 datasets. Neither model shows a substantial advantage for any intermediate layer over its respective best-performing layer.

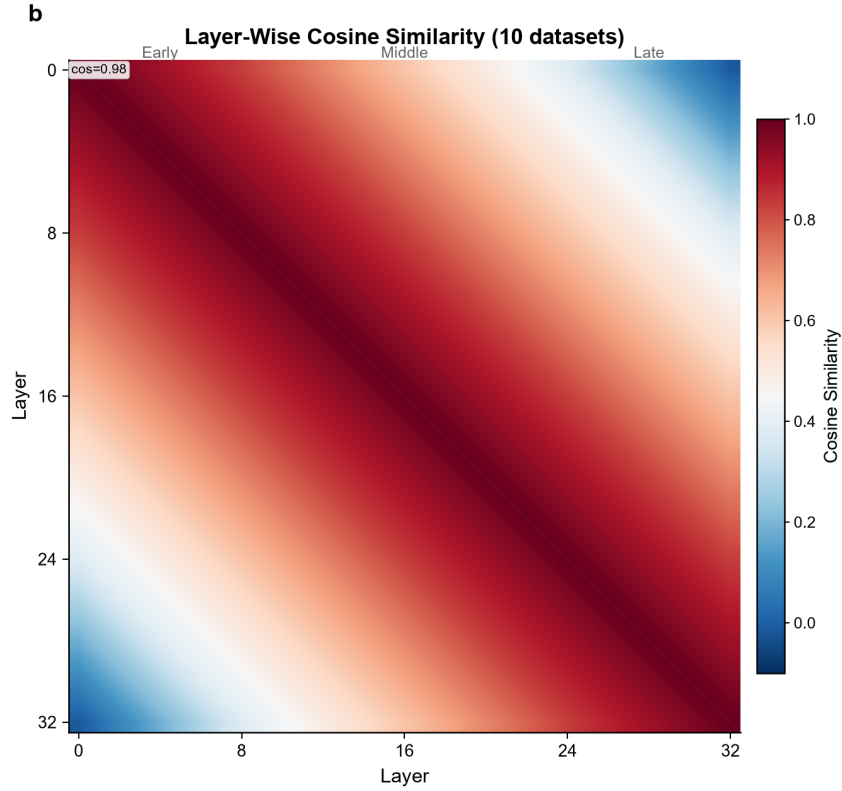


Figure 3: **Layer-wise cosine similarity matrix (33×33)**. Averaged over 10 datasets and 50 random residue positions per dataset. Adjacent layers are highly similar ($\cos = 0.98$), while distant layers approach orthogonality (Layer 0 vs. 32: $\cos = -0.01$). The smooth gradient demonstrates that representations change continuously but information content is conserved.

Layer Invariance in Protein Language Models

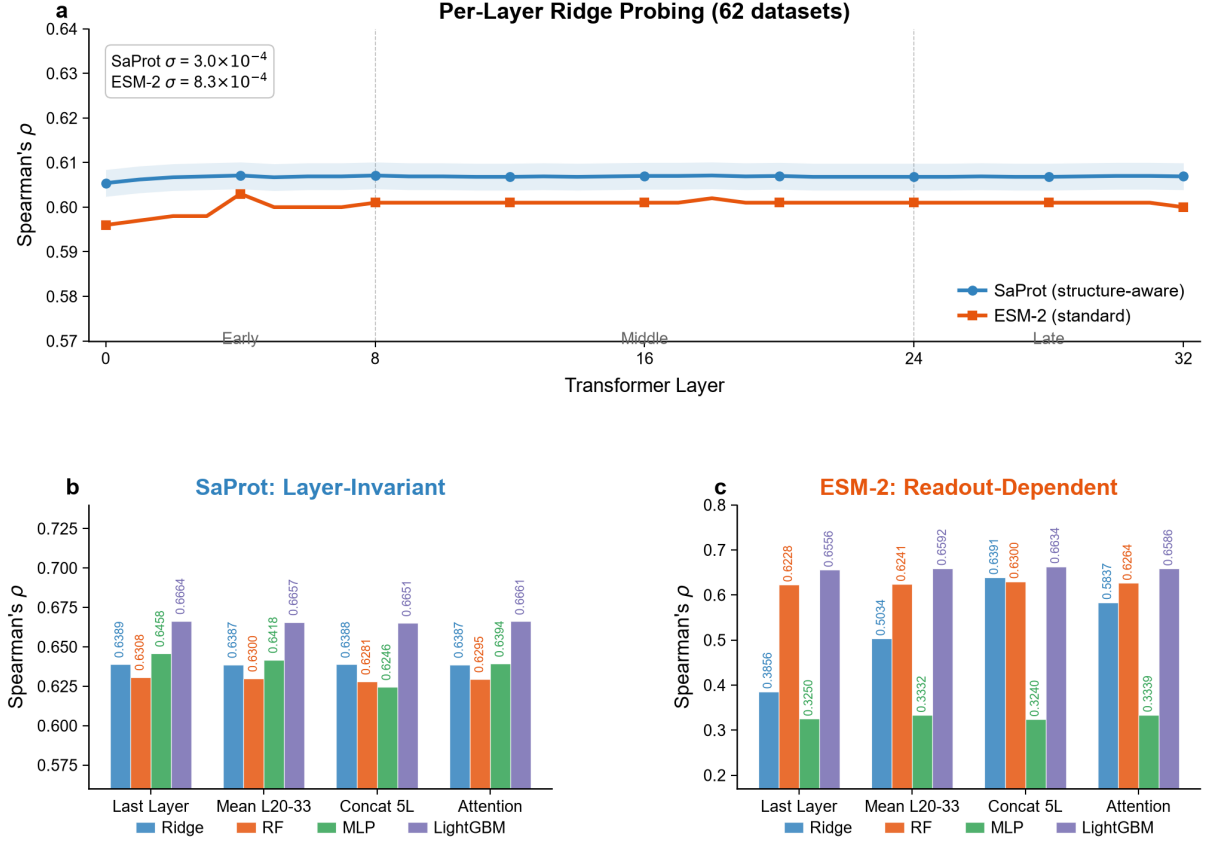


Figure 4: **Per-layer probing and multi-scale pooling: SaProt vs. ESM-2.** (a) 33-layer probing results: all layers within $\Delta\rho < 0.003$ for both models. (b) SaProt pooling (20 datasets): four strategies yield indistinguishable results across Ridge, RF, MLP, and LightGBM readouts; invariance scores $\sigma \leq 2.12 \times 10^{-2}$. (c) ESM-2 pooling (20 datasets): Ridge regression exposes large layer-wise differences (Last Layer $\rho = 0.386$ vs. Concat 5L $\rho = 0.639$), while tree-based readouts (RF ≈ 0.62 – 0.63 , LightGBM ≈ 0.66) eliminate the gap; MLP underperforms across all strategies ($\rho \approx 0.33$).

Random vs. Position-Level Split: Layer Invariance Breakdown

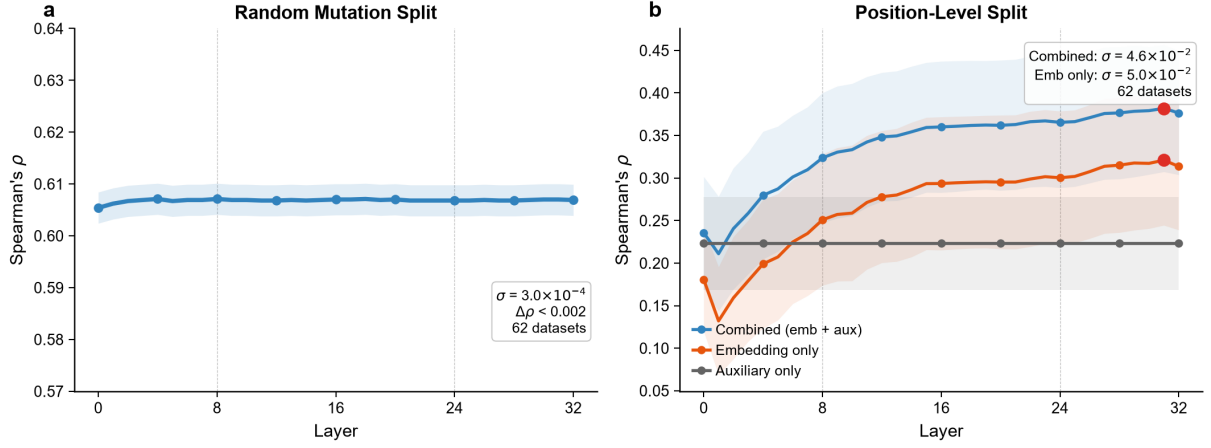


Figure 5: **Position-level vs. random split probing (SaProt, 62 datasets).** (a) Random mutation split: all 33 layers achieve near-identical performance ($\sigma = 3.0 \times 10^{-4}$). (b) Position-level split (GroupKFold on residue positions): overall performance drops by 37–43%, invariance score increases 150-fold ($\sigma = 4.6 \times 10^{-2}$), and the best layer shifts from Layer 8 to Layer 31, indicating that deeper layers capture more position-generalizable features.